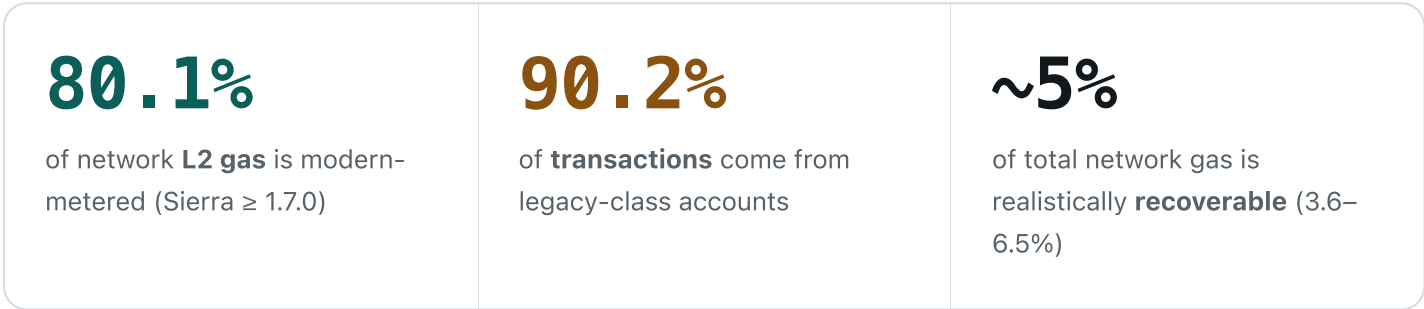


Two accounting systems, one whale, and where the gas actually hides

Starknet meters execution in one of two modes — modern **Sierra-gas** or legacy **Cairo-steps** — decided by the sender account's class version. Here is how mainnet splits between them, and how much gas is actually recoverable.

Sample **2026-07-01** (full UTC day) **1,725** blocks **10,463** txs **265.5B** L2 gas

Source **Starkscan** archive RPC



THE SPLIT

Gas says modern. Transaction count says legacy. Both are true.

The same population, weighted two ways, inverts completely. A few enormous modern transactions carry the gas; a long tail of small legacy transactions carries the count.



- Modern · Sierra-gas (class ≥ 1.7.0)
- Legacy · Cairo-steps (< 1.7.0 + Cairo0)
- Unresolved

WHY MODERN WINS THE GAS COUNT

It isn't "modern accounts." It's one account.

A single sender — the **Extended** perpetuals-DEX operator, 0x048ddc...441f (Sierra 1.7.0) — consumed **78.1% of all network gas** in just **480 transactions**. Its batch `multi_trade` settlements average **432M gas each** — **~74× a normal transaction**. It is the entire "modern-gas" story.

Extended operator — 1 account · 78.1%

all other ~2,900
senders · 21.9%

Strip that one operator out and the picture flips. Among *everyone else*, gas is overwhelmingly legacy — the same story the transaction count already told.

WHOLE NETWORK — GAS

80% / 19%

modern / legacy

EXCLUDING EXTENDED — GAS

9% / 88%

modern / legacy (of the remaining 21.9%)

WHAT EACH SIDE IS ACTUALLY DOING

A perp DEX on one side; infrastructure keepers on the other

Decoding the calldata: modern gas is one exchange settling itself on-chain. Legacy gas is automated keepers — oracles, VRF, relayers — that never upgraded their sender accounts.

Modern · Extended DEX

→ 0x62da07...b470 (exchange + vault)

FUNCTION	NET GAS
multi_trade	68.7%
invest_in_vault	5.2%
price_tick	2.0%
apply_interests	0.6%
new_position	0.5%
redeem_from_vault	0.2%

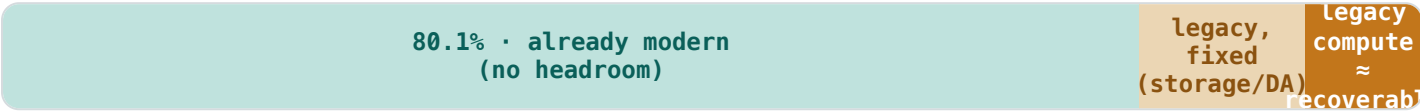
Legacy · Keepers & infra

multiple contracts · bots on Sierra 1.3–1.6

FUNCTION	NET GAS
submit_random · VRF	4.5%
update_oracle · ×3	4.3%
execute_sponsored · AVNU	3.0%
transfer · STRK	0.9%
publish_data_entries	0.8%
attest	0.7%

Modest network-wide, because the biggest consumer is already efficient

Only the *compute* slice of legacy gas re-prices under modern metering — storage, syscalls, data and calldata cost the same either way. Extended (78%) is already Sierra-gas, so the whole headroom lives inside the legacy 19.4%, of which ~55% is compute.



Recoverable ≈ **3.6–6.5% of total network L2 gas** (mid ~5%); absolute ceiling ~9% if all legacy compute vanished. The compute re-prices roughly 2–5× cheaper under Sierra-gas – calibrated from a decomposed AVNU gasless transaction.

WHERE THE LEVERAGE IS

The headroom is concentrated, not diffuse

Legacy gas isn't a fog of retail users — it's a handful of identifiable automated operators still deploying pre-1.7.0 account classes. Each account upgraded captures a disproportionate share.

0x4678eb...61b5	Highest-frequency legacy sender — 5,538 txs/day , 4.4% of network gas. A keeper bot on a Sierra 1.5.0 class.	Sierra 1.5.0
Oracle keepers	<code>update_oracle</code> / <code>publish_data_entries</code> across ≥3 feed contracts — price-push bots, ~5% of gas combined.	< 1.7.0
VRF provider	<code>submit_random</code> — 4.5% of network gas from one randomness operator's account.	< 1.7.0
AVNU relayer	<code>execute_sponsored</code> via the AVNU AA Forwarder — 3.0% of gas, powering gasless UX for others.	< 1.7.0

Method & caveats

MECHANISM

Metering mode is set by the sender account class's Sierra version — `≥ 1.7.0 → Sierra-gas`, else Cairo-steps — and inherited (sticky-downgrade) down the whole call stack. Live since Starknet **v0.13.4** (mainnet ~Mar 2025); constant

`min_sierra_version_for_sierra_gas = 1.7.0`, verified in blockifier source.

DATA

Starkscan archive RPC `getBlockWithReceipts` (sender + L2 gas per tx); class → version via `getClass` · `sierra_program[0:3]`; step decomposition via a v0.7 trace.

CONFIDENCE

Split held across 138 → 752 → 1,725 blocks. Class resolved at chain head (minor skew for accounts upgraded mid-day).

CAVEATS

One-day snapshot; gas is hyper-concentrated, so the 80/20 is essentially one operator's activity. The recovery band is model-based (κ from one decomposed tx) — the hard bounds are legacy = 19.4% and the ~9% ceiling. Time-trend not yet run. ~0.5% of gas unresolved.